

Program : Diploma in Tool And Die Engineering / Manufacturing Technology	
Course Code : 5112A	Course Title: Jigs Fixture And Gauges
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To impart basic knowledge on work holding devices for mass production
- To explain different types of jigs and fixtures used in tool making industry
- To familiarize students with basic aspects of gauging and inspection in modern industry
- To prepare students to design and develop simple jigs and fixtures for a particular part

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Physics		Applied Physics I & II	1 & 2
Mechanics		Engineering Mechanics	2
Production processes		Production engineering I & II	2 & 3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain basic knowledge of jigs and fixtures and apply the concept of degrees of freedom	15	Applying
CO2	Explain the principle of location and clamping	15	Understanding
CO3	Illustrate tool guiding elements and different types of jigs	15	Applying

CO4	Describe various types of fixtures and gauges	13	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2	2					
CO3	3						
CO4	2	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain basic knowledge of jigs and fixtures and apply the concept of degrees of freedom		
M1.01	Express basic concept of work holding devices	3	Understanding
M1.02	Introduce jigs and fixtures	3	Understanding
M1.03	Apply the concept of degrees of freedom of movement to jigs and fixtures	5	Applying
M1.04	Describe general design principle of jigs and fixtures	4	Understanding

Contents:

Work holding concepts – basic work holders, work holder purpose and function, general considerations.

Jigs and Fixtures – introduction- Difference between Jigs and Fixtures – Advantages of jigs and Fixtures - Elements of Jigs and Fixtures – Fool Proofing – Materials used in Jigs and Fixtures

Degrees of Freedom – 12 degrees of freedom – 6 point location principle – (or) 3-2-1 principle of location (simple numerical problems)

Essential features of Jigs and Fixtures – General Design Principles – Design steps– Common defects in Jigs design

CO2	Explain the principle of location and clamping		
M2.01	Identify locating of workpiece in a jig and fixture	4	Understanding
M2.02	Explain various types of locators	4	Understanding
M2.03	Define clamping principle	3	Understanding
M2.04	Explain various clamping methods	4	Understanding
	Series Test - I	1	
Contents:Principles of location – location point – types of locators – Cylindrical – adjustable-long locator- v locator (both fixed and sliding) - conical- diamond pin locator. Six point location of a three legged object – location of a cylinder on a v block-dowel pins. Principles of clamping – types of clamps – lever clamp – hinged clamp – two way clamp – Swinging clamp – wedge clamp- index plate – eccentric clamping arrangement – Quick action clamp – Cam operated clamp - Quarter turn screw – Toggle clamp – Pneumatic and Hydraulic clamps – Washers- ‘C’ Washer – Spherical and flat Washers.			
CO3	Illustrate tool guiding elements and different types of jigs		
M3.01	Acquire basic knowledge about tool guiding elements	3	Understanding
M3.02	Explain various types of bushings	4	Understanding
M3.03	Describe different types of Jigs	3	Understanding
M3.04	Apply theoretical knowledge to design simple jig design	5	Applying
Contents: Tool guiding elements -Jig Bushings: Materials for Jig bushing – press fit bushings – Fixed renewable bushings – slip renewable bushings – liner bushings – screw bushings - miscellaneous type of drill bushings – bushing specifications. Types of Jigs -Drill Jigs: Open drill jig – plate drill jig – box type- template drill jig – channel drill jig – turn over drill jig – angle plate drill jig – closed box drill jig – leaf drill jig – Post jig – Pot jig - indexing drill jig – sandwich jig- universal drill jig. Jig design example – plate jig design.			
CO4	Describe various types of fixtures and gauges		
M4.01	Describe different types of Fixtures	4	Understanding
M4.02	Familiarize simple fixture design	3	Understanding
M4.03	Explain different types of gauges	3	Understanding
M4.04	Recognize standards for gauges	3	Understanding

	Series Test - II	1	
Contents:Type of fixtures – milling fixture-lathe fixture-broaching fixture- grinding fixture- assemble fixture-welding fixture-modular fixture Fixture design example- plain milling fixture Gauges- Limit Gauges – GO and NO GO gauges; types of limit gauges. Gauge design - Taylor's principle of gauging; Gauge tolerance, disposition of gauge tolerance, wear allowance. Three basic type of limit gauges– plug gauge – snap gauge – ring gauge – thickness and length gauges – recess gauges – step gauges – position gauges and receiver gauges			

Text / Reference

T/R	Book Title/Author
T1	K. Venkataraman, Design of Jigs, Fixtures and Press Tools, John Wiley
T2	Kempster, "Introduction to Tool Design and Jigs and Fixtures", ELBS
T3	Balachandran V, Design of Jigs, Fixtures and Press Tools
R1	ASTME, "Hand book of Fixture Design"
R2	S K Hajra Choudhury, Workshop technology vol II
R3	Korsakov, "Fundamental of Fixture Design", MIR Publication, Moscow
R4	Goroshkin.A.K. "Jigs and Fixtures Handbook", MIR Publication, Moscow
R5	P.H.Joshi "Jigs & Fixtures".

Online Resources

Sl.No	Website Link
1	http://www.anyimeasuring.com/
2	https://www.theengineerspost.com/jigs-and-fixtures/
3	http://www.digimat.in/nptel/courses/video/112105126/L35.html
4	https://nptel.ac.in/course.html